Biotic Interaction Detection from Scientific Literature: Knowledge Distillation and Multi-task Learning with Named Entity Recognition

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Abstract

Automated extraction of biotic interactions from scientific literature enables large-scale knowledge graph construction over biodiversity corpora. With Biodiversity PMC (Pasche et al., 2023) alone indexing over 85 million files, systematic manual curation is not feasible. We present a multi-task transformer classifier combining three components: a two-stage teacher chain in which a 122-billion-parameter local model (Qwen3.5-122B; Qwen Team 2025) annotates ∼44,000 harvested sentences with binary labels and a trained ensemble subsequently generates calibrated soft probability labels for knowledge distillation; a multi-task BiomedBERT architecture that jointly classifies sentences and annotates biological entities (HOST, PATHOGEN, SPECIES, INT) using automatically derived BIO supervision; and a warm-start initialization strategy that preserves fine-tuned encoder representations and trains the named entity recognition (NER) head jointly from epoch one; for this architecture and NER scheme, we find that NER pre-training epochs are counterproductive when task-tuned weights are reused. On a 500-sentence multi-source test set spanning five evaluation sources, the classifier achieves F1 = 0.874 (P = 0.925, R = 0.829; 95% bootstrap confidence interval (CI) [0.843, 0.902]). Relative to an identical architecture trained with hard cross-entropy labels, the full pipeline adds +0.202 F1 (p<0.001, McNemar's test), of which +0.162 F1 comes from soft-label distillation alone (cold-start comparison). This gain is largely calibration, not ranking: the same comparison moves the area under the ROC curve (AUC) by only +0.024, indicating soft labels chiefly correct the probability scale rather than the model's underlying discrimination. The remaining +0.040 F1 (p=0.002) comes jointly from warm-start initialization and skipping NER pre-training — neither is sufficient alone (Section 3, Finding 2). The proposed model surpasses the BiomedBERT × FLAN-T5-base ensemble teacher (F1 = 0.850) at 3.3× lower inference cost and is statistically indistinguishable from the stronger single-task template-trained baseline (F1 = 0.875, p=0.760) while additionally producing NER output. On real-literature sources alone (excluding synthetic), the proposed model leads the template-trained baseline (F1 = 0.847 vs. 0.821), indicating the aggregate tie is driven by the synthetic subset. The classifier is deployed upstream of BiotXplorer (Ruch et al., 2024), the SIBiLS (Gobeill et al., 2020) semantic search platform for biotic interactions.

1 Introduction

Taxon interactions shape ecosystem structure, drive disease dynamics, and determine the response of ecological communities to perturbation. Cataloguing these interactions requires synthesising knowledge distributed across millions of scientific publications. The Global Biotic Interactions database (GloBI; Poelen et al. 2014) aggregates over 13 million structured interaction records from more than 700 tabular data sources, yet covers only a fraction of the interaction types documented across the biodiversity literature, with 23 of 171 ROBI relation types (ROBIext v2025, GloBI's relation ontology; Poelen et al. 2014) represented in our experiments. The SIBiLS platform (Gobeill et al., 2020), the Swiss Institute of Bioinformatics Literature Services, provides federated semantic search over Biodiversity PMC (Pasche et al., 2023), a curated corpus of over 85 million biomedical and biodiversity documents encompassing both PubMed Central and MEDLINE content.

Automated information extraction from scientific text has a well-established tradition in the biomedical domain. Named entity recognition (NER) and relation classification methods have been applied to protein–protein interactions (Bunescu et al., 2005), drug–drug interactions (Lim et al., 2018), and gene–disease associations, with deep learning substantially improving entity detection accuracy (Habibi et al., 2017). A widely used strategy for generating labelled training data at scale is distant supervision: known interaction databases are aligned with text to produce labels automatically, without exhaustive manual annotation (Mintz et al., 2009). These approaches are predominantly designed for molecular and clinical literature; ecological interaction types (predation, parasitism, pollination, herbivory, mutualism) span a broader taxonomic scope and a more heterogeneous publication landscape, with direct consequences for model generalisation (Thessen et al., 2012).

A particular challenge in literature-based interaction mining is the co-occurrence problem: a sentence may mention two organism names and an interaction term together without actually describing an interaction between them. Co-occurrence-based retrieval achieves high recall but suffers from substantial false-positive rates (Bunescu et al., 2005); sentence-level classifiers that discriminate true interactions from co-occurrence artefacts are a standard component of biomedical text-mining pipelines, and the extracted triples can populate biodiversity knowledge graphs supporting disease spread, food web, and conservation reasoning (Page, 2016; Poisot et al., 2021). Knowledge gaps remain large, particularly for field-observed interactions published in ecology journals outside the PubMed index (Poisot et al., 2021).

We address this gap with three primary contributions:

1. A two-stage teacher chain combining large language model (LLM) binary annotation with ensemble-based soft probability distillation, yielding a 44K-sentence training corpus with calibrated labels.
2. A multi-task BiomedBERT architecture with automatic NER auxiliary supervision; a warm-start ablation reveals that initializing from a fine-tuned BiomedBERT encoder and omitting NER pre-training surpasses the two-phase cold-start approach (+0.040 F1).
3. A quantitative distribution mismatch analysis identifying the interaction types responsible for the current performance ceiling and guiding future data collection.

2 Methods

2.1 Task Definition

Given a sentence s from scientific literature, predict $y \in \{0, 1\}$ where $y = 1$ indicates that s describes a biotic interaction between two or more named organisms with the following interaction types in scope: predation, parasitism, pathogen infection, pollination, herbivory, mutualism, symbiosis, seed dispersal, competition, and vector-borne transmission. The Relation Ontology provides the conceptual framework of our study.

2.2 Training Corpus Construction

Sentence sources. Two independent sentence corpora are used for different training purposes: a synthetic template corpus that trains the single-task baseline, and a real-literature harvested corpus that trains the multi-task champion via knowledge distillation (Section 2.4).

Template-derived sentences (synthetic). Structured triples from GloBI (e.g. *parasitizes(Anisakis simplex, Gadus morhua)*) are injected into handwritten sentence templates covering ten interaction categories. Every positive sentence was individually validated by a large language model to remove formulaic or linguistically implausible candidates, yielding 25,081 clean sentences (7,251 positive).

PMC-harvested sentences (real). Articles co-mentioning known GloBI taxon pairs were queried from Biodiversity PMC (Pasche et al., 2023), both by taxon co-occurrence and by interaction category via the SIBiLS API (Gobeill et al., 2020). Co-occurrence sentences were extracted and labeled by a local Qwen3.5-122B model (a 122-billion-parameter checkpoint in the Qwen3 model family described by Qwen Team 2025) using the prompt: *"Does this sentence describe a biotic interaction between two named organisms? Answer yes or no only."* Of 44,178 harvested sentences, 4,065 (9.2%) were accepted as positives.

Soft label generation. The accepted 44,178 sentences were re-scored by the trained BiomedBERT $\times$ FLAN-T5-base ensemble, producing a soft probability $p_{\text{ensemble}} \in [0, 1]$ per sentence. These soft labels constitute the distillation teacher signal (Section 2.4).

2.3 Model Architecture Evolution

Our experiments are split and evaluated using five distinct steps with an increasing order of complexity.

Classical baselines. A support vector machine (SVM) and Random Forest on term frequency-inverse document frequency (TF-IDF) features achieved F1 $\approx$ 0.62, establishing the need for contextual representations.

Template-trained BiomedBERT. We use the BiomedBERT model (Gu et al., 2021) (`microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract-fulltext`), a BERT-base model (110M parameters, 12 transformer layers, 768-dimensional hidden states) pre-trained from scratch on PubMed abstracts and PubMed Central full-text articles. A linear classification head on the [CLS] token representation is fine-tuned using 5-fold cross-validation, AdamW optimiser ($\eta = 2 \times 10^{-5}$, weight decay = 0.01), and threshold optimisation on the validation split.

All subsequent models (ensemble, distilled student, multi-task champion) use this same BiomedBERT encoder and differ only in training objective or auxiliary supervision. This baseline (Table 2, row 6) is trained on template-generated training data via standard cross-entropy and achieves F1 = 0.875 on the 500-sentence test set. Its strong aggregate performance masks divergent per-source behaviour (Figure 1), discussed in detail in Section 4.2: neither this baseline nor the multi-task warm-start champion (Section 2.5) dominates uniformly across all five test sources. This model also acts as the BiomedBERT component of the teacher ensemble described below.

BiomedBERT $\times$ FLAN-T5-base ensemble. FLAN-T5-base (Chung et al., 2022), a sequence-to-sequence model trained with instruction tuning, was fine-tuned as a binary classifier by prompting for a yes/no answer and scoring the first generated token. The geometric mean of BiomedBERT and FLAN-T5-base predicted probabilities, with joint threshold optimisation, achieves F1 = 0.850 on the 500-sentence test set. The gain over a single model is attributable to model complementarity: BiomedBERT handles precise, template-like language while FLAN-T5-base generalises better to paraphrased ecological sentences.

The two subsequent stages, knowledge distillation and multi-task NER, are described in Sections 2.4 and 2.5 respectively. Figure 4 (Appendix E) summarises the F1 progression across architecture stages shown numerically in Table 2.

2.4 Knowledge Distillation

Knowledge distillation (Hinton et al., 2015) transfers calibrated uncertainty from a teacher to a student by training on soft probability distributions rather than hard labels (Buciluă et al., 2006), which is especially valuable for borderline sentences.

Training objective. The student BiomedBERT is trained using Kullback-Leibler (KL) divergence against the ensemble teacher:

$$\mathcal{L}_{\text{KL}} = T^2 \cdot \text{KL}(\sigma(\mathbf{z}_s/T) \parallel \sigma(\mathbf{z}_t/T)) \quad (1)$$

where T is the temperature, $\mathbf{z}_s$ and $\mathbf{z}_t$ are student and teacher logits, and σ denotes the softmax function. Rows without soft labels use hard cross-entropy. The combined loss is $\mathcal{L} = \alpha \cdot \mathcal{L}_{\text{KL}} + (1-\alpha) \cdot \mathcal{L}_{\text{CE}}$.

Optimal hyperparameters (T=2, α=0.5) were identified by grid search over six student configurations (Appendix A). Temperature T=2 spreads the teacher distribution sufficiently to transfer calibrated uncertainty without over-smoothing; it also corrects for ensemble overconfidence at p>0.9 (expected calibration error, ECE = 0.148 on EP-A), where the ensemble's mean confidence of 0.967 exceeds its accuracy of 0.804. The best-performing student achieves F1 = 0.858 on the 500-sentence test set.

2.5 Multi-task BiomedBERT with NER Auxiliary Task

Architecture. The BiomedBERT encoder is extended with two linear output heads that share all encoder weights:

- *Classification head*: reads the [CLS] token vector ($\mathbb{R}^{768} \to \mathbb{R}^2$), produces P(interaction).
- *NER head*: reads every token vector ($\mathbb{R}^{768} \to \mathbb{R}^{|\mathcal{L}|}$), produces Beginning-Inside-Outside (BIO; Ramshaw and Marcus 1995) tagging-scheme label logits.

NER label scheme. We index $\approx$4.2 million binomial taxon names plus 21K common names into a gazetteer, matched against text via Aho-Corasick automata (Aho and Corasick, 1975) for exact string lookup.

We use the `full_typed` scheme with $|\mathcal{L}| = 9$ labels: `O` (background), `B/I-HOST` (hosting organism), `B/I-PATHOGEN` (infecting or parasitic agent), `B/I-SPECIES` (ambiguous role in symmetric interactions), `B/I-INT` (interaction verb or phrase). In addition to taxon gazetteers, labeling relies on an interaction lexicon developed within

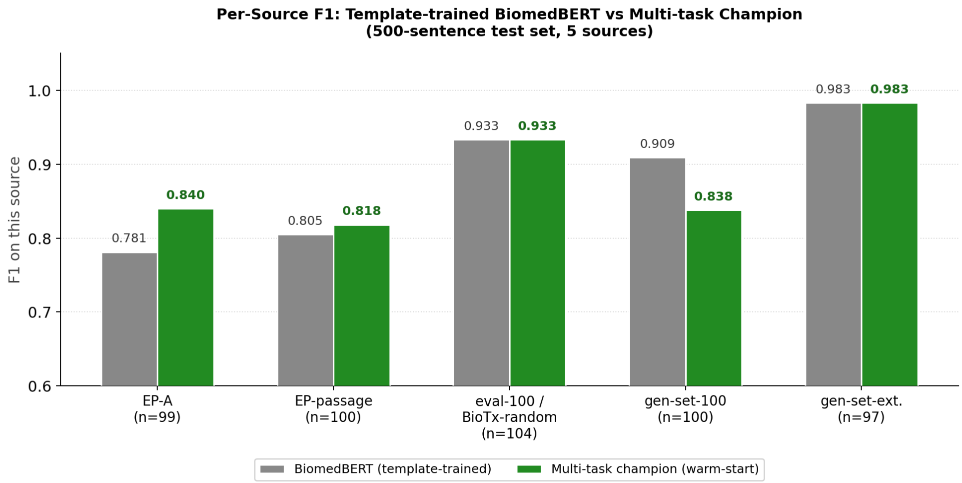

Figure 1: Per-source F1 comparing the template-trained BiomedBERT baseline against the multi-task warm-start champion (Section 2.5). Per-source n shown on the x-axis; "gen-set-ext." is gen-set-150-extension (Section 2.6), 97 sentences kept after independent gold-label validation out of 150 drafted.

SIBiLS (Gobeill et al., 2020), containing 591 terms derived from the GloBI relation ontology (ROBI), covering all interaction types indexed by BiotXplorer, supplemented by 50 biomedical terms for pathogen-host vocabulary not represented in ROBI. Gold taxon annotations from training metadata override gazetteer lookup when available.

Table 1 compares the four label schemes evaluated. The HOST/PATHOGEN distinction encodes the direction of the relation: who acts and who is acted upon. This directional labelling is motivated by the downstream knowledge graph construction task, where edge direction is semantically significant.

Training strategy. We compare two initialization strategies.

Cold-start (two-phase). When initializing from general pretrained BiomedBERT weights, training proceeds in two phases designed to avoid interference between the NER and classification objectives (Liu et al., 2019):

Phase 1: NER pre-train (E_{NER} epochs). The classification head is frozen; only the encoder and NER head receive gradient updates from token-level cross-entropy loss $\mathcal{L}_{\text{NER}}$, priming the encoder to locate taxon names and interaction verbs first.

Phase 2: Joint training (E_{joint} epochs). All parameters are unfrozen. Combined loss:

$$\mathcal{L} = \alpha \cdot \mathcal{L}_{\text{cls}} + (1-\alpha) \cdot \mathcal{L}_{\text{NER}}$$

where $\mathcal{L}_{\text{cls}}$ is soft KL divergence against ensemble soft labels when available, and hard CE otherwise. With $E_{\text{NER}}=2$ and $E_{\text{joint}}=3$, this achieves F1 = 0.835 (Table 2, row 2).

Warm-start (proposed champion). We initialize the BiomedBERT encoder from the Template-trained BiomedBERT classification checkpoint and skip Phase 1 entirely: both the NER head and the classification head train jointly from epoch one. This strategy achieves F1 = 0.874 at $\tau=0.360$, the best result in our evaluation. Ablation over the number of NER pre-training epochs under warm-start reveals a clear negative correlation (Appendix B, warm-start rows): F1 = 0.874 with 0 NER epochs, 0.860 with 1, and 0.822 with 2. We attribute this to *representation disruption*: the template-trained encoder already encodes biotic interaction patterns from classification fine-tuning, and NER pre-training updates it toward entity-detection representations instead, degrading the classification signal. Warm-starting without Phase 1 preserves this initialization while the NER head still learns entity recognition jointly in Phase 2.

2.6 Evaluation Protocol

All models are evaluated on a 500-sentence test set (281 positive, 56.2%) combining five sources: EP-A (99 sentences, GloBI-seeded), EP-passage (100 sentences, GloBI-seeded), a merged eval-100/BioTx-random source (104 sentences), gen-set-100 (100 synthetic sentences), and gen-set-150-extension (97 additional synthetic sentences restoring the test set to 500). Source construction and validation details are in Section 2.7 and Appendix D. We report precision, recall, and F1-score. The decision threshold τ is optimised on the validation

Table 1: NER label schemes compared in the configuration analysis (Section 3; B = beginning, I = inside span). F1 is the best *cold-start* result per scheme (full ablation in Appendix B). `full_typed` lacks the highest cold-start F1 but is carried to the warm-start ablation (0.874, Table 2) for its HOST/PATHOGEN typing (Section 4.3).

Scheme	# labels	Label set	F1
basic	**3**	**O, B/I-SPECIES**	**0.861**
`typed`	5	O, B/I-HOST, B/I-PATHOGEN	0.794
`full`	5	O, B/I-SPECIES, B/I-INT	0.858
`full_typed`	9	O, B/I-HOST, B/I-PATHOGEN, B/I-SPECIES, B/I-INT	0.840

Table 2: Main results on the 500-sentence test set (281 positives). The proposed warm-start multi-task model (top, **champion**) significantly outperforms the hard-CE ablation (+0.202 F1, p<0.001) and surpasses the ensemble teacher at 3.3× lower inference cost (a single BiomedBERT forward pass versus the ensemble's additional FLAN-T5-base sequence-to-sequence decoding step). Full configuration sweep in Appendix B. F1 values are rounded to three decimals for display; reported deltas and significance tests use unrounded scores, so pairwise differences may not equal the difference of displayed values.

Model	F1	P	R	AUC
Multi-task BiomedBERT (`full_typed`, α=0.5, warm-start, 0 NER ep)	**0.874**	**0.925**	**0.829**	**0.918**
Multi-task BiomedBERT (`full_typed`, α=0.5, cold-start, 2 NER ep)	0.835	0.885	0.790	0.881
Same arch., hard cross-entropy (no distillation)	0.673	0.920	0.530	0.857
Distilled student (BiomedBERT, T=2, α=0.5)	0.858	0.877	0.840	0.920
Ensemble BiomedBERT × FLAN-T5-base	0.850	0.925	0.787	0.912
Template-trained BiomedBERT (single-task baseline)	0.875	0.881	0.868	0.921

split, not the test set. For the warm-start champion, $\tau = 0.360$ is derived from the standard 15% held-out split of its training data. For the cold-start ablation, $\tau = 0.140$ is derived from the same kind of split; this lower value reflects the cold-start model's probability distribution skewing toward lower scores (Appendix C).

Bootstrap confidence intervals are estimated with 10,000 resamples (Efron and Tibshirani, 1993). Pairwise significance is assessed using McNemar's test (McNemar, 1947) with continuity correction.

2.7 Test Set Integrity

We ran three checks (exact-match, near-duplicate, taxon-pair leakage, Appendix D) between the test set and every training corpus used to produce a reported model. These checks confirm the test set is independent of the training data used to produce every reported result, with one exception: exactly one test sentence (from a SIBiLS-harvested source) also appears in a broader corpus not used to train any reported model. The same procedure uncovered that the eval-100 and BioTx-random sources, nominally independent, share 96 of 100 sentences as the same underlying content exported in two formats; they are merged into a single deduplicated source (Section 2.6). The checks cannot rule out overlap with BiomedBERT's own pretraining corpus (PubMed abstracts and full-text articles, Gu et al. 2021), since several test sources are themselves drawn from PMC and we have no access to that corpus to verify this either way. A related gold-label provenance caveat, distinct from sentence-level leakage, is discussed in Section 4.5.

3 Results

3.1 Multi-task Configuration Analysis

Table 2 compares the champion configuration against key baselines; the full sweep over all NER schemes, α values, initialization strategies, and pretraining epochs is given in Appendix B. AUC, reported alongside F1/P/R as a threshold-independent measure of ranking quality, broadly agrees with the F1 picture: the champion, template-trained baseline, distilled student, and ensemble form a tight cluster (AUC 0.912–0.921) with overlapping discriminative power, while hard-CE trails (0.857). Notably, hard-CE's AUC gap to this cluster (0.055–0.064) is much smaller than its F1 gap (0.177–0.202), consistent with the calibration argument in Section 2.4: hard-CE's underlying ranking ability is closer to the soft-label models than its F1 alone suggests, and its larger weakness is poor probability calibration at the operating threshold.

Four consistent findings emerge from the analysis:

1. *Soft labels are the largest single lever (+0.162 F1).* The `hardce` configuration is architecturally identical to the cold-start champion (`full_typed` scheme, α=0.5, 2 NER pre-training epochs, cold-start) and differs only in the classification loss (hard cross-entropy instead of soft-KL), achieving F1 = 0.673. The soft-KL variant (cold-start multi-task) reaches 0.835: a difference of +0.162 F1 (p<0.001, McNemar's test) attributable purely to the loss function, larger than any architectural change. The further +0.040 F1 to reach the warm-start champion (0.874) is a separate, architectural effect (warm-start initialization with zero NER pre-training epochs), isolated in Finding 4 below.

2. *NER pre-training epochs matter only under warm-start.* Under cold-start, 0 and 2 NER pre-training epochs are statistically indistinguishable (0.840 vs. 0.835). Under warm-start, the picture is unambiguous: adding epochs *hurts* (0 ep: 0.874 > 1 ep: 0.860 > 2 ep: 0.822). NER pre-training disrupts the fine-tuned encoder's classification representations when they already exist; with no such representations under cold-start, epoch count matters far less.

3. *NER scheme richness alone does not predict cold-start performance.* Under cold-start (all four schemes trained on identical data), there is no monotonic trend by label-set size: `basic` (0.861) and `full` (0.858) outperform `full_typed` (0.840), while `typed` (0.794), HOST/PATHOGEN-only, is weakest of all. `full_typed`'s advantage materializes only with warm-start (Finding 4), reaching 0.874, the highest result overall; we retain it regardless because its HOST/PATHOGEN typing is required for the downstream knowledge-graph use case (Section 4.3).

4. *Warm-start initialization closes the gap with the template-trained baseline.* Initializing from the template-trained BiomedBERT checkpoint and skipping NER pre-training achieves F1 = 0.874, an improvement of +0.040 over the cold-start champion (0.835). The gap with the template-trained single-task model (F1 = 0.875) is 0.0003 and is not statistically significant (Section 3.2). The warm-start champion additionally produces NER labels at no extra inference cost.

3.2 Statistical Significance

McNemar's test (McNemar, 1947) is appropriate for this comparison because predictions are binary (correct or incorrect per sentence), the models are evaluated on the same test examples (paired observations), and the test specifically targets prediction disagreements rather than assuming a continuous performance metric. Under the null hypothesis that two models commit errors on the same sentences, the reported p-value is the probability of observing prediction disagreements as extreme as those measured.

Figure 2 shows 95% bootstrap confidence intervals on the 500-sentence test set.

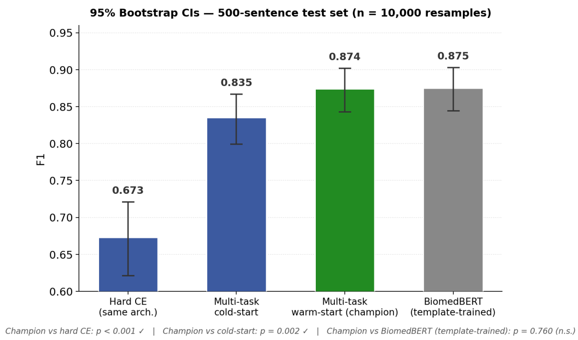

Figure 2: Bootstrap 95% confidence intervals for four models on the 500-sentence test set. Significance annotations refer to McNemar's test.

The warm-start champion significantly outperforms the hard cross-entropy baseline (p<0.001): 113 sentences correctly classified by the champion but not by hard CE, versus 35 in the reverse direction (+0.202 F1). It also significantly outperforms the cold-start multi-task model (p=0.002): 32 sentences correctly classified by warm-start but not by cold-start, versus 11 in the reverse direction (+0.040 F1). Compared with the single-task template-trained baseline (F1 = 0.875), the champion's deficit is 0.0003 F1 and is not statistically significant (p = 0.760, 23 vs. 20 disagree in each direction): the multi-task warm-start model is statistically indistinguishable from the template-trained baseline while additionally providing NER output. By contrast, the cold-start multi-task model is significantly weaker than the template-trained baseline (p=0.045), underscoring that warm-start initialization, not the multi-task architecture alone, is what closes this gap. The 95% bootstrap CIs for the champion [0.843, 0.902] and the template-trained baseline [0.844, 0.903] overlap substantially, confirming reliable model selection between them is not possible at this test set size (see also Section 4.5).

Threshold sensitivity and operating-point trade-offs are analysed in Appendix C.

3.3 Distribution Mismatch

The training corpus, harvested via the SIBiLS API, is dominated by predation (42%), herbivory (22%), and pollination (22%) sentences. Parasitism/Host interactions represent only 10% of training positives, and Pathogen/Infection only 3%, yet together these two types constitute 68% of test positives (Parasitism/Host 41.6%, Pathogen/Infection 26.3%), visualised in Figure 5 (Appendix E).

This mismatch reflects a structural property of the harvested corpus: clinical and biomedical pathogen papers dominate its open-access content, while field-observed host-parasite interactions remain underrepresented. The full 500-sentence test set spans five annotation sources with diverse interaction-type distributions, including synthetic sentences covering all ten in-scope categories (gen-set-100). The distributional gap between training and test is expected: it identifies where to invest in future data collection to improve robustness across interaction types.

4 Discussion

4.1 Soft Labels as the Primary Driver of Improvement

The soft-label and warm-start gains quantified in Section 3 (Finding 1) are consistent with the knowledge distillation literature (Hinton et al., 2015; Tang et al., 2019): a teacher exposed to diverse linguistic expressions of interactions assigns calibrated probabilities to borderline sentences, encoding semantic uncertainty hard labels cannot represent. This is a calibration mechanism rather than a ranking improvement (Section 3): soft labels act mainly on *where* the decision boundary falls, not on the model's ability to separate positives from negatives.

4.2 Matching Template-Trained Performance via Warm-Start

The warm-start multi-task champion is statistically indistinguishable from the template-trained baseline (Section 3.2). Figure 1 breaks this aggregate equivalence down by test source: the champion is more competitive on the two GloBI-seeded passage sources (EP-A: 0.840 vs. 0.781; EP-passage: 0.818 vs. 0.805), the template-trained baseline is stronger on gen-set-100 (0.909 vs. 0.838), and the two models are within 0.001 F1 of each other on the eval-100/BioTx-random and gen-set-150-extension sources. The aggregate tie is therefore not a uniform tie at the source level, but two models with complementary strengths reaching the same overall score.

Grouping by provenance sharpens this further: the champion leads on the 303 real-literature sentences (directional, p=0.067), while the template-trained baseline leads on the 197 synthetic sentences (0.946 vs. 0.913) — the aggregate tie is the net of two opposing effects, not genuine equivalence.

By contrast, the cold-start champion (F1 = 0.835) is nominally weaker than the template-trained baseline (p=0.045); with five comparisons reported here, this would not survive a Bonferroni correction (α=0.01), so we treat it as suggestive only. The key difference is that cold-start training on the 44K corpus displaces the template-aligned representations built during template pretraining, whereas warm-start retains them. The warm-start champion additionally outputs NER labels (HOST, PATHOGEN, SPECIES, INT, Finding 4), unlike the template-trained baseline's binary-only output.

4.3 NER as Zero-Cost Auxiliary Supervision

The NER auxiliary task improves classification without additional manual annotation: BIO labels (Ramshaw and Marcus, 1995) come from taxon gazetteers and interaction lexicons (distant supervision, Mintz et al. 2009), and the NER head produces entity labels at inference time at no extra cost. The HOST/PATHOGEN distinction it outputs directly encodes knowledge-graph edge direction, benefiting both BiotXplorer's search ranking and downstream relation extraction. Extended discussion of the mechanism and concrete operational use cases is given in Appendix E.

4.4 The Training Distribution Ceiling

Given that the architecture-level gap with the template-trained baseline is already closed (Section 4.2), and the training/test distribution mismatch identified in Section 3 is large relative to it, further improvement is more likely to come from expanding the training corpus's coverage of parasitism/host and pathogen/infection interactions than from further architectural changes.

4.5 Limitations

1. *Teacher chain dependence.* The soft-label ensemble was trained on template-derived

and early PMC-harvest data (pre-Qwen), so Qwen3.5-122B and the ensemble are orthogonal signal sources; overlap with the template baseline's training data is zero, and only 5.1% of distillation sentences appear elsewhere in the corpus (predominantly low-confidence negatives, mean $p_{\text{ensemble}} = 0.20$). Soft labels are also imperfectly calibrated (Section 2.4). The same dependence touches test-gold validation: gen-set-150-extension's labels were validated by the same Qwen3.5-122B model that generates every training label, sharing a common annotator bias for that source. EP-A is human-curated and independent of both teachers, our clean anchor against this risk.

2. *Test set composition.* The 500-sentence test set combines five sources, two synthetic (gen-set-100, gen-set-150-extension). No single style uniformly favours either model (Section 4.2). 95% CIs span ± 0.030 F1. Marginal CIs overlap for the champion–baseline pair, so they cannot be separated on F1 alone; the paired McNemar test still resolves the larger pairwise gaps reliably (Section 3.2).
3. *Binary classification.* The classifier detects interactions but does not classify interaction type. Downstream relation extraction is needed to produce typed knowledge graph edges.
4. *NER head not directly evaluated.* We evaluate the NER head only through its effect on classification; direct entity-span evaluation (HOST/PATHOGEN/SPECIES/INT) against manually curated gold is left to future work, as the distant-supervision labels share the gazetteer's blind spots and cannot serve as an independent gold standard.
5. *English only.* Training and evaluation are exclusively English-language.

5 Conclusion

We have presented a multi-task BiomedBERT classifier for sentence-level biotic interaction detection that achieves F1 = 0.874 (P = 0.925, R = 0.829) on a 500-sentence multi-source test set. The model is built on three components: a two-stage teacher chain combining large-LLM binary annotation with ensemble soft-probability distillation over 44K sentences; a multi-task NER auxiliary task with automatic BIO supervision from taxon gazetteers; and a warm-start initialization from a fine-tuned BiomedBERT encoder that trains the NER head jointly from epoch one.

The central quantitative findings are: (1) soft-label distillation is the largest single lever, contributing +0.162 F1 over hard cross-entropy on identical architecture and data ($p<0.001$), though the matching AUC change of +0.024 shows this gain is predominantly improved calibration at the operating threshold rather than better ranking; (2) warm-start initialization combined with skipping NER pre-training adds a further +0.040 F1 ($p=0.002$) — neither component suffices alone — bringing the total gain over hard CE to +0.202 F1; and (3) the warm-start multi-task model is statistically indistinguishable from the template-trained baseline (F1 = 0.875, $p=0.760$) while surpassing the ensemble teacher (F1 = 0.850) at 3.3× lower inference cost and additionally providing NER output for downstream knowledge graph construction.

The classifier is deployed in BiotXplorer (`https://biotxplorer.sibils.org/`) as an upstream quality filter and constitutes the first component of a knowledge graph construction pipeline for biodiversity and host-pathogen interactions. Future work will address: (1) harvesting from non-PubMed ecological literature to expand training coverage of parasitism and host-interaction types; (2) typed relation extraction to produce directed knowledge graph triples; (3) integration with amplicon sequencing pipelines (MetaPlantCode project; Gardette et al. 2024), where interaction knowledge can support ecological coherence re-ranking of taxon lists derived from environmental DNA samples.

Ethics Statement

This work raises no ethical concerns. Training data consists of sentences automatically extracted from open-access scientific literature and labeled by automated models; no personal data was collected or used. The classifier is designed to support literature mining for biodiversity knowledge graph construction and carries no foreseeable potential for misuse.

Data and Code Availability

Training code, evaluation scripts, and the test set are available at `https://anonymous.4open.science/r/biotic-interaction-classifier-anon-28DC`. Model weights can be reproduced from the provided training configuration.

References

Alfred V Aho and Margaret J Corasick. 1975. Efficient string matching: An aid to bibliographic search. *Communications of the ACM*, 18(6):333–340.

Cristian Buciluă, Rich Caruana, and Alexandru Niculescu-Mizil. 2006. Model compression. In *Proceedings of the 12th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, pages 535–541.

Razvan Bunescu, Ruifang Ge, Rohit J Kate, Edward M Marcotte, Raymond J Mooney, Arun K Ramani, and Yuk Wah Wong. 2005. Comparative experiments on learning information extractors for proteins and their interactions. *Artificial Intelligence in Medicine*, 33(2):139–155.

Hyung Won Chung, Le Hou, Shayne Longpre, Barret Zoph, Yi Tay, William Fedus, Eric Li, Xuezhi Wang, Mostafa Dehghani, Siddhartha Brahma, and 1 others. 2022. Scaling instruction-finetuned language models. *arXiv preprint arXiv:2210.11416*.

Bradley Efron and Robert J Tibshirani. 1993. *An introduction to the bootstrap*. Chapman & Hall.

Aurélien Gardette, Yanis Sklab, Eugeni Belda, Eric Chenin, and Jean-Daniel Zucker. 2024. META-PLANTCODE: Harmonizing plant metabarcoding pipelines in europe. *Biodiversity Information Science and Standards*, 8:e135729.

Julien Gobeill, Déborah Caucheteur, Pierre-André Michel, Luc Mottin, Emilie Pasche, and Patrick Ruch. 2020. SIB literature services: RESTful customizable search engines in biomedical literature, enriched with automatically mapped biomedical concepts. *Nucleic Acids Research*, 48(W1):W12–W16.

Yu Gu, Robert Tinn, Hao Cheng, Michael Lucas, Naoto Usuyama, Xiaodong Liu, Tristan Naumann, Jianfeng Gao, and Jinfeng Peng. 2021. Domain-specific language model pretraining for biomedical natural language processing. *ACM Transactions on Computing for Healthcare*, 3(1):1–23.

Maryam Habibi, Leon Weber, Mariana Neves, David L Wiegandt, and Ulf Leser. 2017. Deep learning with word embeddings improves biomedical named entity recognition. In *Bioinformatics*, volume 33, pages i37–i48.

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. Distilling the knowledge in a neural network. In *NIPS Deep Learning and Representation Learning Workshop*.

Sangrak Lim, Kyubum Lee, Kihyun Choi, and Jaewoo Kang. 2018. Drug-drug interaction extraction from the literature using a recursive neural network. *PloS ONE*, 13(1):e0190926.

Xiaodong Liu, Pengcheng He, Weizhu Chen, and Jianfeng Gao. 2019. Multi-task deep neural networks for natural language understanding. In *Proceedings of ACL*, pages 4487–4496.

Quinn McNemar. 1947. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157.

Mike Mintz, Steven Bills, Rion Snow, and Dan Jurafsky. 2009. Distant supervision for relation extraction without labeled data. In *Proceedings of ACL-IJCNLP*, pages 1003–1011.

Roderic D M Page. 2016. Towards a biodiversity knowledge graph. *Research Ideas and Outcomes*, 2:e8767.

Emilie Pasche, Julien Gobeill, Donat Agosti, Lyubomir Penev, Quentin Groom, Teodor Georgiev, Esteban Gaillac, Alexandre Flament, Déborah Caucheteur, Pierre-André Michel, and Patrick Ruch. 2023. From SIBiLS to biodiversity PMC: Foundations for the One Health Library. *Biodiversity Information Science and Standards*, 7:e111660.

Jorrit H Poelen, James D Simons, and Chris J Mungall. 2014. Global biotic interactions: An open infrastructure to share and analyze species-interaction datasets. *Ecological Informatics*, 24:148–159.

Timothée Poisot, Gabriel Bergeron, Kevin Cazelles, Tad Dallas, Dominique Gravel, Andrew MacDonald, Benjamin Mercier, Clément Violet, and Steve Vissault. 2021. Global knowledge gaps in species interaction networks data. *Journal of Biogeography*, 48(7):1552–1563.

Qwen Team. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.

Lance A Ramshaw and Mitchell P Marcus. 1995. Text chunking using transformation-based learning. In *Third Workshop on Very Large Corpora*.

Patrick Ruch, Emilie Pasche, Pierre-André Michel, and Déborah Caucheteur. 2024. BiotXplorer: Navigating evidence-based biotic interactions. *Biodiversity Information Science and Standards*, 8:e135453.

Raphael Tang, Yao Lu, Linqing Liu, Lili Mou, Olga Vechtomova, and Jimmy Lin. 2019. Distilling task-specific knowledge from BERT into simple neural networks. *arXiv preprint arXiv:1903.12136*.

Anne E Thessen, Hong Cui, and Dmitry Mozzherin. 2012. Applications of natural language processing in biodiversity science. *Advances in Bioinformatics*, 2012:391574.

A Knowledge Distillation Hyperparameter Search

Table 3: Full distillation hyperparameter grid on the 500-sentence test set (281 positives). These are single-task distilled classifiers (no NER head, not the multi-task architecture used elsewhere in this paper), so their calibration and threshold τ=0.25 (training-split-derived) are independent of, and not comparable to, the multi-task champion's τ=0.360 or the cold-start ablation's τ=0.140. All variants use BiomedBERT as student except v4 (DistilBERT) and v5 (SciBERT). Rows marked ‡ use checkpoints no longer available for re-evaluation on the current test set version; their values are retained from a prior test-set version for completeness.

Student	T	α	**Test F1**
v1 BiomedBERT	4	0.7	0.851
v2 BiomedBERT (best)	**2**	**0.5**	**0.858**
v3 BiomedBERT‡	4	0.9	0.704†
v4 DistilBERT‡	2	0.5	0.785
v5 SciBERT‡	2	0.5	0.799
v6 BiomedBERT (T=1.5)‡	1.5	0.5	0.783

†v3 predicted positive for nearly all sentences at τ=0.25 on the prior, more balanced test-set version (R = 1.000, P = 0.510); AUC = 0.880 confirmed it was calibrated but threshold τ=0.25 was misaligned with that class balance.

B Full Multi-task Configuration Analysis

Table 4: All multi-task configurations evaluated on the 500-sentence test set (281 positives). NER pt = NER pre-training epochs before joint training. Init = encoder initialization (cold = pretrained BiomedBERT; warm = Template-trained BiomedBERT checkpoint). All configurations use soft labels except `hardce`. Thresholds are training-split-derived per config. Rows marked ‡ use checkpoints from an earlier exploration phase that are no longer available for re-evaluation on the current test set version; their values are retained from a prior test-set version for completeness and should not be compared directly to the unmarked rows.

Config	**NER scheme**	α	**Init**	**NER pt**	**F1**	**P**	**R**	**AUC**
Cold-start configurations								
basic_a05	basic	0.5	cold	0	0.861	0.884	0.840	0.905
typed_a05	typed	0.5	cold	0	0.794	0.897	0.712	0.890
full_a03	full	0.3	cold	0	0.858	0.892	0.826	0.903
full_typed_a05	full_typed	0.5	cold	0	0.840	0.879	0.804	0.891
full_a05_ner2‡	full	0.5	cold	2	0.781	0.859	0.717	—
full_typed_a03_ner2‡	full_typed	0.3	cold	2	0.802	0.834	0.772	—
full_typed_a05_ner2	full_typed	0.5	cold	2	0.835	0.885	0.790	0.881
full_typed (5 joint ep)‡	full_typed	0.5	cold	2+5	0.798	0.836	0.764	—
Warm-start configurations (encoder from template-trained BiomedBERT)								
full_typed_a05 (warm, 0 NER ep)	full_typed	**0.5**	warm	**0**	**0.874**	**0.925**	**0.829**	**0.918**
full_typed_a05 (warm, 1 NER ep)	full_typed	0.5	warm	1	0.860	0.881	0.840	0.910
full_typed_a05 (warm, 2 NER ep)	full_typed	0.5	warm	2	0.822	0.902	0.754	0.898
hardce (cold, 2 NER ep)	full_typed	0.5	cold	2	0.673	0.920	0.530	0.857
distilled student (no NER)	—	—	cold	—	0.858	0.877	0.840	0.920
Ensemble BERT×T5	—	—	—	—	0.850	0.925	0.787	0.912

C Threshold Analysis

Figure 3 shows precision, recall, and F1 as a function of the decision threshold for an earlier cold-start checkpoint on a 300 positive + 700 negative held-out validation split, kept here to illustrate the general shape of the threshold–metric relationship, which is qualitatively similar across checkpoints. The cold-start and warm-start configurations reported in Table 2 use freshly-trained checkpoints with thresholds derived from each model's own standard 15% held-out validation split (τ=0.140 and τ=0.360 respectively, Section 2.6), not from the 300+700 split shown below.

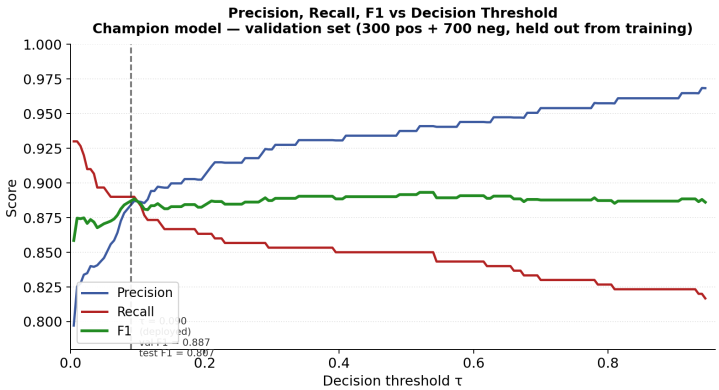

Figure 3: Precision, recall, and F1 versus decision threshold on a 300 positive + 700 negative held-out validation split (not seen during training) for an earlier cold-start checkpoint, illustrating threshold-curve shape rather than the currently-reported operating points. The annotated threshold τ=0.090 achieves F1 = 0.887 on this validation split. The currently-reported champion (Table 2) uses τ=0.360, derived from its own 15% held-out validation split, achieving test F1 = 0.874.

Both champions assign soft probabilities via knowledge distillation: genuine interactions score predominantly in the range 0.2–0.6 rather than near 1.0. For the checkpoint shown, F1 is nearly flat across $\tau \approx 0.09$–0.92 on the validation split (0.887–0.893), confirming that threshold selection is robust.

For applications where false positives entering a knowledge base are costly, operating at higher τ increases precision at the cost of recall. For the cold-start ablation: P = 0.885, R = 0.790 at τ=0.140. For the warm-start champion (P = 0.925, R = 0.829 at τ=0.360), raising τ further prioritises precision over recall analogously. The threshold curves make these trade-offs explicitly visible and allow practitioners to choose the operating point appropriate to their downstream task.

D Test Set Integrity Details

This appendix gives the full methodology behind the summary in Section 2.7: three checks between the test set and every training corpus used to produce a reported model.

Exact-match. Comparing normalised (lower-cased, whitespace- and punctuation-collapsed) test sentences against the 44,178-sentence distillation corpus (champion training data) and the 25,081-sentence template corpus (single-task baseline training data) yields zero exact matches in both cases. Against the broader 34,880-sentence corpus used only for the teacher-chain analysis in Section 4 (not used to train any reported model), exactly one test sentence is also present, originating from a SIBiLS-harvested source. This same exact-match procedure additionally uncovered that the eval-100 and BioTx-random sources, nominally independent, share 96 of 100 sentences as the same underlying content exported in two formats; they are merged into a single deduplicated source (Section 2.6) rather than double-counted. The 97 gen-set-150-extension sentences added to restore the test set to 500 were authored fresh for this evaluation and are by construction absent from all training corpora.

Near-duplicate. Using TF-IDF character n-gram (3–5) cosine similarity with a threshold of 0.85, no additional near-duplicates are found beyond the single exact match identified above.

Taxon-pair leakage. The three GloBI-seeded test sources (EP-A, EP-passage, BioTx-random) are seeded from specific taxon pairs; even where the exact sentence differs, the same pair could also seed a training template, constituting a subtler form of leakage. Comparing each source's 100 taxon pairs against the template corpus's pairs shows overlap of 1% (EP-A), 1% (EP-passage), and 0% (BioTx-random).

E Extended Discussion: Architecture Figures and Downstream Utility

E.1 Utility for BiotXplorer and Knowledge Graph Construction

The classifier provides two concrete operational benefits for BiotXplorer. First, as an upstream filter, it eliminates sentences that merely co-mention taxa without describing an interaction, reducing index noise and improving search relevance. Second, the calibrated probability output allows the BiotXplorer

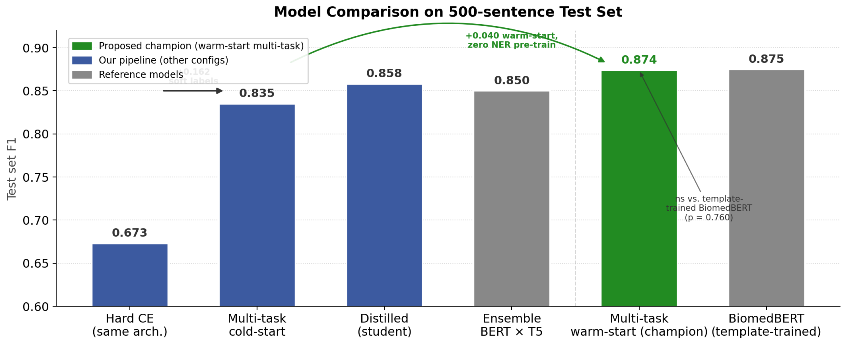

Figure 4: Test F1 across model architecture stages (numerically in Table 2). Annotated arrows indicate the primary driver of each improvement.

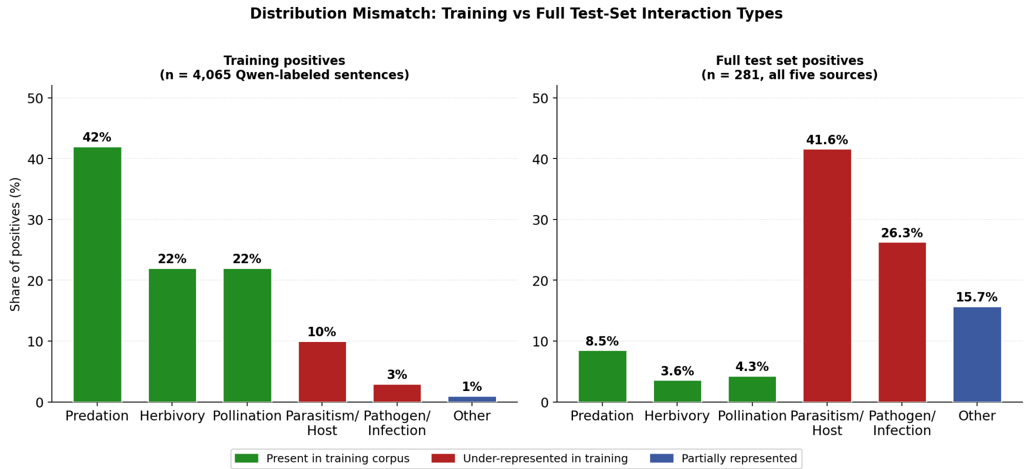

Figure 5: Distribution of interaction types in training positives (left, n=4,065 Qwen-labeled sentences) versus the full test set's positives (right, n=281, aggregated across all five annotation sources). Categories are classified from GloBI interaction terms (EP-A, EP-passage, BioTx-random/eval-100) and the category field of gen-set-100 and gen-set-150-extension. Colors match between panels to highlight the same categories.

platform to rank results by interaction confidence rather than simple keyword match, enabling more nuanced retrieval.

For downstream knowledge graph construction, the multi-task model's NER labels provide additional structure: the HOST/PATHOGEN distinction produced by the NER head directly encodes the direction of the knowledge graph edge (who interacts with whom), enabling the downstream relation extractor to assign typed edges rather than undirected links. This is particularly relevant for pathogen-host and parasite-host pairs where edge direction is biologically meaningful.